



Periparturient sequence of physiology surrounding dyscalcemia in multiparous Holstein dairy cows: A prospective cohort study

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ABSTRACT

Although most cows effectively normalize their calcium status following the physiological decline in serum total calcium (tCa) that occurs after parturition, a substantial proportion of dairy cows develop dyscalcemia. Dyscalcemia is defined as tCa ≤ 2.2 mmol/L at 4 DIM and is associated with poor productivity, reduced reproductive success, and increased risk of disease. The physiological relationships underlying these associations are poorly understood; therefore, our objective in this prospective cohort study was to establish the sequence of physiology related to dyscalcemia. We enrolled 45 multiparous Holsteins and followed them from 14 d prepartum through 14 DIM. Daily DMI and rumination times were recorded for the duration of the study period. We collected blood samples at –14, –7, –3, and –1 d relative to parturition, every 8 h from parturition through 96 h postpartum, and again at 120 h, 168 h, 240 h, and 336 h postpartum. Serum samples were analyzed for tCa, haptoglobin, and BHB concentrations. Milk yields were evaluated from every milking through 336 h postpartum, and daily milk yields averaged by week were collected through 10 wk of lactation. Cows were classified as dyscalcemic (DYS; tCa ≤ 2.2 mmol/L; $n = 12$) or eucalcemic (EU; tCa > 2.2 mmol/L; $n = 33$) based on tCa at 96 h postpartum. Linear mixed effects models were fit for each physiological measure modeled as a function of dyscalcemia status and its interaction with time, accounting for relevant confounding effects and including the random effect of cow. Although we did not detect differences between DYS and EU cows prepartum for any of our measures, we observed marked differences in temporal patterns of postpartum physiological indicators associated with dyscalcemia. Dyscalcemic cows experienced lower intake starting between 24 h and 48 h postpartum (DYS: mean = 16.3 kg/d, 95% CI = 14.5–18.1 kg/d; EU: mean = 20.6 kg/d,

95% CI = 19.5–21.7 kg/d) and lower rumination times beginning at 40 h (DYS: 25 min/h, 95% CI = 19–31 min/h; EU: 38 min/h, 95% CI = 35–42 min/h), followed by higher haptoglobin concentrations at 64 h (DYS: 0.69 g/L, 95% CI = 0.49–0.98 g/L; EU: 0.39 g/L, 95% CI = 0.32–0.48 g/L), and finally lower tCa concentration at 96 h (DYS: 2.11 mmol/L, 95% CI = 2.02–2.21 mmol/L; EU: 2.36 mmol/L, 95% CI = 2.31–2.43 mmol/L). Milk production was lower for dyscalcemic cows throughout 14 DIM by 1.4 kg (95% CI = 0.6–2.1 kg) on average per milking, and milk production was also lower for dyscalcemic cows through 10 wk in lactation by 4.8 kg/d (95% CI = 2.3–7.4 kg/d) on average. Although our study may have limited generalizability due to its single farm and seasonality setting, we have demonstrated that under controlled conditions, early postpartum maladaptation associated with dyscalcemia is first apparent through depressed feed intake and rumination shortly after calving. This sequence of physiology may be relevant to inform the development of interventions.

Key words: calcium, inflammation, intake, rumination

INTRODUCTION

As a dairy cow transitions from gestation into the early postpartum period, she enters a potentially fragile moment in her productive cycle. During a few short days, she must successfully coordinate an adaptive balancing act to restore tissue homeostasis in the reproductive tract and maintain homeorhesis in the face of hormonal changes, nutrient deficits, altered immune functionality, inflammatory activation, and exposure to microbial threats (Bauman and Currie, 1980; Goff and Horst, 1997; Brewer et al., 2020). Failure to adapt appropriately to these metabolic and immunological demands can result in impaired health and performance (Ingvarsen et al., 2003; LeBlanc et al., 2005; Carvalho et al., 2019; Macmillan et al., 2021).

Dyscalcemia is a calcium-related phenotype of multiparous Holstein cows whereby calcium status is depressed at 4 DIM (total serum calcium concentration [tCa] ≤ 2.2

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

mmol/L; Neves et al., 2018; McArt and Oetzel, 2023). This phenotype is a useful indicator that cows may be failing to adapt to the demands of the early postpartum period, as dyscalcemic cows are known to experience a variety of negative outcomes, including increased risk of disease, reduced reproductive efficiency, and reduced milk production (McArt and Neves, 2020; Seely and McArt, 2023; Seminara et al., 2023). Increased systemic inflammation during the early postpartum period has also been associated with maladaptive outcomes (Huzzey et al., 2009; Nightingale et al., 2015; McCarthy et al., 2016) and is hypothesized to be a central driver in this system (LeBlanc, 2012; Bradford et al., 2015). Dyscalcemia and calcium status in general are thought to be intertwined with inflammation, but in situ research designed to assess this relationship is currently insufficient (Couto Serrenho et al., 2025; Seminara et al., 2025). Both dyscalcemia (Seely et al., 2021a; Seely and McArt, 2024) and inflammatory activation (Trevisi et al., 2015; Cocco et al., 2021) have been independently associated with lower DMI and rumination times, but the temporal sequence by which these conditions develop concurrently during the periparturient period remains poorly understood. Furthermore, it is well known that reduced DMI can result in ketone body formation, and that increased concentrations of ketones in the blood can be a negative health indicator, but where early postpartum ketone physiology fits into this complex puzzle is still a mystery (McArt et al., 2012; Eduardo Rico and Barrientos-Blanco, 2024).

The historical paradigm around the early postpartum period has been that parturition precipitates changes in calcium regulation, intake, and nutrient metabolism, and that these changes lead to immunosuppression and a variety of inflammatory processes that go on to impair reproductive efficiency and milk production (Bauman and Currie, 1980; Curtis et al., 1983; Goff and Horst, 1997). Despite the foundational research that contributed to this paradigm, it has recently been under scrutiny because of inconsistency in the designs and findings of key studies (Horst et al., 2021). Induced inflammatory models have been cited as evidence to support these challenges to traditional thinking (Horst et al., 2020; Ogenorth et al., 2024a), but holistic evaluations of causal architecture and temporal sequences of physiological responses during the immediate postpartum period are sparse.

Therefore, our primary objective in this prospective cohort study was to assess the temporal patterns of periparturient physiology from 14 d before calving to 336 h postpartum (14 DIM), in cows with and without dyscalcemia, to understand the sequence by which physiological dysregulation takes place. Serum haptoglobin concentrations were of principal interest in our design as a measure of physiological activity, although other measures of physiology included DMI, rumination times,

tCa, BHB concentrations, and early postpartum milk production. We hypothesized that reductions in DMI would begin immediately following parturition in dyscalcemic cows, followed by elevated inflammation, ketone production, calcium perturbances, depressed rumination, and ultimately losses in milk production, in that order. Our secondary objective was to evaluate milk production between dyscalcemic and eucalcemic cows on a longer time scale (through 10 wk in milk) to confirm the long-term relevance of early postpartum physiological dysregulation, and we hypothesized that milk production would be impaired in dyscalcemic cows through 10 wk.

MATERIALS AND METHODS

Our prospective cohort study was designed and ambidirectionally analyzed following the STROBE-Vet reporting guidelines (Sargeant et al., 2016).

Sample Size Calculation

We conducted our sample size calculation using R version 4.5.1 in RStudio (Posit Software, PBC, Boston, MA) with the intention to detect a 0.2 g/L difference in haptoglobin concentrations immediately following calving, with SD of 0.15 g/L (Huzzey et al., 2009). Controlling for both type I and type II errors at 5% ($\alpha = 0.05$ and power = 0.95) and estimating dyscalcemia incidence of 30% (McArt and Neves, 2020; Seely et al., 2021a), we arrived at a target sample size of 40 cows. We inflated this sample size by 20% to account for potential exclusions due to gestation length <260 d and the occurrence of clinical hypocalcemia within the first 24 h postpartum. With this inflation, our target sample size was 48 cows.

Study Population

Our study was conducted at the Cornell University Ruminant Center (Harford, NY) from July through November 2024. All eligible cows ($n = 48$) were enrolled the day before dry-off upon exiting the milking parlor after the 1400 h (midday) milking. Cows were not eligible for enrollment if they were lame, had 3 teats, or were fistulated. Dry cows were housed in a freestall barn with access to pasture during the day. Cows were moved once weekly, at an average of -20 d (± 1 d) relative to expected calving, into individual tiestalls bedded with mattresses and kiln-dried sawdust, where they remained through 336 h postpartum. After 336 h postpartum, cows were moved to a lactating pen in a freestall barn until they left the study at 70 DIM. After transitioning into lactation, cows were milked 3 times daily. The time of calving was recorded for each cow and broadly categorized as morning (0600–1400 h), afternoon (1400–2200 h), or night (2200–0600 h).

Feed Sampling, Analysis, and Measurement of DMI and Rumination

Rations were formulated using the Cornell Net Carbohydrate and Protein System v. 6.5.5 (AMTS.Cattle.Professional v. 4.17.0.0, AMTS LLC; Van Amburgh et al., 2015). Feed was delivered once daily to the dry cow freestall barn at 0900 h. All dry cows were fed a diet formulated to have a negative DCAD for the entirety of the dry period; however, cows between dry-off and –20 d before expected calving also had daily access to pasture. Once cows were moved to the tiestall barn, urine pH was tested 3 times weekly on >33% of all cows that had been in the barn for at least 3 d, to check the extent to which the diet was effectively acidifying cows, in accordance with common milk fever prevention protocols (Goff, 2008).

Dry cows in the tiestalls were fed once daily at 0800 h at individual feed bins, from which orts were collected and weighed on a daily basis. Daily intake was calculated by subtracting the weight of the day's orts from the weight of feed offered the previous day; thus the intake reported on any given day was the total intake from the previous 24-h period. Prepartum intake on the day of calving was excluded from analysis, as each cow calved at a different time of day and therefore had a different and incomplete 24-h period to access dry cow feed.

After calving, cows were immediately transitioned to an early-lactation diet and a 3-times-per-day feeding protocol, whereby feed was delivered at 0600 h, 1400 h, and 2200 h, and orts from the previous 8-h period were collected, weighed, and fed back. Additional feed was supplied at each feeding, if necessary, to allow for ad libitum intake. Intake for each 24-h period following calving was calculated from the sum of intakes during each 8-h feeding period, such that the total DMI at 24 h postpartum was the sum of intakes at 8 h, 16 h, and 24 h postpartum. The 3-times-per-day feeding protocol began at the earliest of the feeding times possible following parturition. This protocol continued to 120 h postpartum, at which point cows were transitioned back to a once-a-day feeding schedule through 336 h postpartum. During both the pre- and postpartum periods, feed delivery was adjusted daily to target 10% refusal while allowing for ad libitum intake.

Samples of TMR were collected weekly and dried in a forced-air oven at 40°C for 96 h to determine DM. Dry matter intake was calculated by multiplying the raw intake by the percent dry matter of the feed being fed at each time point. Four-week composites of dried samples were created once monthly and ground through a 2-mm screen using a Wiley mill (Arthur H. Thomas Co.). Each composite TMR was analyzed for nutrient composition using wet chemistry at a commercial laboratory (Cum-

Table 1. Analyzed nutrient composition (% DM $\pm$ SD unless otherwise noted) of pre- and postpartum diets fed to multiparous Holstein cows (n = 45) during the periparturient period

Composition, % DM	Diet	
	Prepartum	Postpartum
CP	13.9 $\pm$ 0.1	18.2 $\pm$ 0.3
ADF	28.9 $\pm$ 1.0	19.9 $\pm$ 0.5
aNDF	43.6 $\pm$ 1.0	30.3 $\pm$ 0.5
Starch	15.6 $\pm$ 0.8	25.9 $\pm$ 0.3
NFC	32.7 $\pm$ 0.9	40.0 $\pm$ 2.2
Fat	3.2 $\pm$ 0.4	4.1 $\pm$ 0.4
Ca	1.39 $\pm$ 0.05	0.85 $\pm$ 0.08
P	0.31 $\pm$ 0.00	0.36 $\pm$ 0.01
Mg	0.47 $\pm$ 0.01	0.39 $\pm$ 0.01
K	0.97 $\pm$ 0.01	1.36 $\pm$ 0.03
S	0.44 $\pm$ 0.01	0.26 $\pm$ 0.01
Na	0.15 $\pm$ 0.00	0.57 $\pm$ 0.10
Cl	0.56 $\pm$ 0.03	0.45 $\pm$ 0.09
DCAD, mEq/100 g of DM	–11.8 $\pm$ 1.3	31.1 $\pm$ 2.2

berland Valley Analytical Services, Waynesboro, PA); wet chemistry results are shown in Table 1.

Rumination times (RT; min/h) were recorded on every cow every 2 h using validated neck collar rumination loggers (SCR, Merck and Co., Rahway, NJ; Schirmann et al., 2009). Data were aligned based on the precise time of parturition and aggregated to 8-h measurements by taking the average of all 2-h measurements recorded during the previous 8-h period.

Blood Sample Collection and Analysis

Using 20-gauge Vacutainer needles and 10-mL evacuated anticoagulant-free tubes (Becton Dickinson), blood samples were collected from coccygeal vessels at a range of times throughout the study period. Cows were first sampled at the time of enrollment the day before dry-off after their afternoon milking. Dry cows were sampled once weekly at the time of morning feed delivery until –20 d ($\pm$ 1 d) relative to their expected calving. Once moved into the tiestall barn, cows were sampled 3 times per week at the time of morning feed delivery until parturition. This sampling strategy resulted in the cohort being sampled for blood at –14 d ($\pm$ 0.7 d), –7 d ($\pm$ 0.6 d), –3 d ($\pm$ 0.5 d), and –1 d ($\pm$ 0.3 d) relative to parturition. From parturition through 96 h postpartum, cows were sampled once every 8 h after milkings and feed deliveries (0630 h, 1430 h, and 2230 h). Cows were sampled again at 120 h, 168 h, 240 h, and 336 h postpartum. Only samples taken between –14 d relative to parturition and 336 h postpartum were used in this analysis. Samples were allowed to clot for 30 min at room temperature, and serum was separated via centrifugation at 2,000 $\times$ g for 15 min at 20°C. Multiple 1.5-mL aliquots were stored at –20°C until laboratory analysis.

One aliquot of serum was analyzed for tCa using commercially available kits (Ca Gen.2, Roche Diagnostics) at the New York State Animal Health Diagnostic Center (Ithaca, NY) on an automated analyzer (Hitachi Modular P800, Roche Diagnostics). The inter- and intra-assay coefficients of variation for this analysis were 1.2% and 1.0%, respectively. Haptoglobin was assessed from a second aliquot at the Animal Health Laboratory (Guelph, Ontario, Canada) on an automated analyzer (Cobas c501, Roche Diagnostics) using a method based on Makimura and Suzuki (1982). Inter- and intra-assay coefficients of variation were 7.0% and 2.3% respectively. A third aliquot of serum was used to assess BHB concentrations using a Nova Vet handheld BHB meter (Nova Biomedical Co., Waltham, MA), validated for use with both whole blood and serum with coefficients of variation <10% (Bach et al., 2016; Rodriguez et al., 2021).

Milk Weight Collection

Milk weights were recorded for each milking from calving through 336 h postpartum, every 8 h relative to calving. Colostrum from the first milking, as well as transition milk from the second and third milkings of each cow were collected into buckets and weighed on a scale by hand. For every milking after the third milking, milk weights were collected on an individual basis from milk meters below the parlor (DeLaval, Tumba, Sweden). For longer-term milk production data, daily milk weights, recorded as the sum of all milkings in a day, were collected by the herd management software throughout the study period (DairyComp305, Valley Ag, Tulare, CA). These data were averaged on a weekly basis and collected as weekly averages of daily yield through 10 wk of lactation.

Disease Diagnosis and Treatment

Early postpartum diseases were diagnosed by farm personnel using standard case definitions described subsequently, following physical examination by the farm manager. If a cow developed disease at any point in the postpartum period, they were treated using the farm's standard protocols. No cow experienced a calving that required more than slight assistance (calving ease scores all $\leq 2/5$; López de Maturana et al., 2007; Dhakal et al., 2013). Cows unable to stand, lift their head, or eructate were suspected to have parturient paresis and, in accordance with farm protocol, were treated immediately with intravenous calcium and excluded from the study. No other type of supplemental calcium was administered to study cows for any other reason. Metritis was diagnosed when cows had rectal temperature $\geq 39.5^{\circ}\text{C}$ in conjunction with a red to brown watery, malodorous

uterine discharge. Cows that developed metritis were treated with ceftiofur hydrochloride (Excenel, Zoetis), and, depending on the extent of febrility, might also have been treated with flunixin meglumine (Banamine, Merck Animal Health), at the farm's discretion. Pneumonia was diagnosed when cows had rectal temperature $\geq 39.5^{\circ}\text{C}$ with abnormal respiratory signs, lethargy, and decreased intake, determined following respiratory auscultation and behavioral assessment, and this condition was also treated with ceftiofur hydrochloride. Hyperketonemia was assessed by farm personnel using a handheld BHB meter (BHBCheck Plus, PortaCheck) once during the first week of lactation, and cows with blood BHB >1.2 mmol/L were treated with oral propylene glycol. All therapeutic interventions were applied between the morning and afternoon milkings.

Statistical Analysis

All calculations of descriptive statistics and statistical analyses were performed using R version 4.5.1 in RStudio (Posit Software, PBC, Boston, MA). Descriptive statistics were generated using general functions contained in the base and stats packages of R.

Linear mixed effects models provided in the nlme package (Pinheiro and Bates, 2000; Pinheiro et al., 2025) were used to assess the relationships between dyscalcemia and each outcome of interest. Pre- and postpartum data were modeled separately for each of the response variables using consistent model fitting strategies for each timeframe.

Prepartum DMI, Rumination, tCa, Haptoglobin, and BHB. To evaluate relationships between postpartum dyscalcemia status and prepartum response variables, we started by fitting a maximal model that retrospectively predicted the response as a function of dyscalcemia status (dyscalcemic, **DYS**: tCa ≤ 2.2 mmol/L at 96 h postpartum; eucalcemic, **EU**: tCa >2.2 mmol/L at 96 h postpartum) as a categorical variable; time (d prepartum) as a categorical variable; interaction between dyscalcemia status and time; time of day during which the cow calved (morning, afternoon, or night) and its interaction with time to account for potential diurnal changes in physiology that might be present; cow parity group (2, 3, or ≥ 4) and its interaction with time to account for potential parity level differences in temporal patterns; each cow's previous-lactation 305-d mature-equivalent milk production (**p305ME**) to control confounding due to underlying biological milk production potential at the cow level; and a random effect of cow to account for intra-cow correlations across time points. Using a backward stepwise approach, predictors with $P > 0.1$ were removed from the models to improve parsimony. Dyscalcemia status, time, and their interaction, as our main predictors of in-

terest, along with the known confounders parity group and p305ME were included in every prepartum model regardless of statistical significance.

Postpartum DMI, Rumination, tCa, Haptoglobin, BHB, and Short-Term Milk Yield. To evaluate postpartum patterns of our physiological measures, and short-term milk yield, a very similar model-fitting strategy was employed, whereby dyscalcemia status, time, their interaction, parity group, and p305ME were included in every model. Key differences of the strategy included the measurement of time as hours postpartum, and the inclusion of time of day of calving in every model regardless of statistical significance, due to known differences in diurnal patterns of certain analytes (Seely et al., 2021b). Postpartum models also included a binary variable indicating whether a cow was treated with a medication (ceftiofur hydrochloride, flunixin meglumine, or propylene glycol) following disease diagnosis. Each treatment was included as a separate variable, and variables remained 0 until the hour of treatment, at which point they were set to 1 until the end of the sampling period. Due to unknown potential confounding effects of these treatments, these variables were retained in every model regardless of statistical significance. Four additional variables were tested in each postpartum model: a binary variable indicating whether a cow had twins or an issue occurred during parturition (twins or stillbirth), the weight of the calf as a continuous variable (or the largest calf if twinning occurred), and the interactions between each of these terms and time to account for potential differences in temporal responses of our measures to different calving conditions. Each of these variables were removed from postpartum models if $P > 0.1$. The random effect of cow was included in all models.

Postpartum Long-Term Milk Yield. Average 10-wk milk production was modeled using similar linear mixed effects models. Covariates in this model included dyscalcemia status, time (measured in week postpartum), their interaction, parity group, p305ME, and binary variables for each of the potential early postpartum treatments, regardless of statistical significance. Time of day at calving, calving issue, calf weight, and each of their interactions with time, as well as the interaction of parity group and time, were all tested in this model and removed where $P > 0.1$. A random effect of cow was also included in the final model for this outcome.

Our results are presented as proportionally weighted marginal means with 95% CI calculated from each model, for each calcium status, at each time point with pairwise comparisons using the emmeans package (Lenth, 2025). We have adjusted the results of our pairwise comparisons using the Benjamini-Yekutieli procedure (Benjamini and Yekutieli, 2001) to control the false discovery rate within a multiple comparisons context with dependent data. For

pre- and postpartum models, as well as the longer-term milk production model, the correlation structures autoregressive, compound symmetry, and unstructured were tested for each outcome with equally spaced observations (DMI, rumination, short- and long-term milk production), and in every case the autoregressive structure was selected for the final model based on having the lowest Bayesian information criteria. For serological measures (tCa, haptoglobin, and BHB), due to the uneven spacing of observation times, only compound symmetry and unstructured correlation structures were tested, and for each of the pre- and postpartum final models except prepartum haptoglobin, compound symmetry was selected based on having the lowest Bayesian information criteria, as earlier. For prepartum haptoglobin, the unstructured correlation structure had the lowest Bayesian information criterion. Model fit was assessed using visual inspection of diagnostic plots. A plot of residuals over the range of model predictions was used to establish whether the homoscedasticity assumption was met, and a histogram of residuals was used to establish whether the residual normality assumption was met. If either assumption was not met, outcome variables were transformed using a natural logarithm (postpartum haptoglobin and both pre- and postpartum BHB) for modeling, which resulted in both assumptions being satisfied and measures being back-transformed for presentation.

RESULTS

Descriptive Characteristics

Of the initial 48 cows enrolled in our study, 1 cow with previous days carried calf <260 was excluded ($n = 1$). Additionally, 2 cows developed parturient paresis, which required immediate intravenous calcium supplementation, and both were excluded from the study ($n = 2$). These exclusions left 45 cows eligible for the final analysis.

Within the final cohort ($n = 45$), 12 (27%) were DYS and 33 (73%) were EU. Descriptive characteristics of categorical and continuous data by calcium dynamic group are presented in Tables 2 and 3, respectively. The mean prepartum urine pH was 5.7 (95% CI = 5.5–5.9). Cefotiofur hydrochloride treatment was administered to all cows diagnosed with metritis (DYS: $n = 2$; EU: $n = 3$) and 1 cow with suspected pneumonia (DYS: $n = 1$), at an average DIM of 7 (± 3 DIM). The 2 cows with metritis that also received flunixin meglumine received this treatment at 3 or 4 DIM (DYS: $n = 1$; EU: $n = 1$). Oral propylene glycol solution was administered to all cows developing hyperketonemia at any point during the first 14 DIM (DYS: $n = 4$; EU: $n = 8$), and this therapy was applied at an average DIM of 6.5 (± 1.3 DIM).

Table 2. Descriptive characteristics of categorical variables and disease events for 45 multiparous Holstein cows by calcium status group; calcium status group classification was based on serum total calcium (tCa) at 96 h postpartum (dyscalcemic, DYS: tCa ≤ 2.2 mmol/L; eucalcemic, EU: tCa > 2.2 mmol/L)

Item	Calcium status group	
	EU (%)	DYS (%)
Total cows, n (% ¹)	33 (73)	12 (27)
Parity group, n (% ²)		
2	9 (27)	3 (25)
3	11 (33)	3 (25)
≥ 4	13 (39)	6 (50)
Time of parturition, n (% ²)		
Morning	13 (39)	5 (42)
Afternoon	11 (33)	3 (25)
Night	9 (27)	4 (33)
Health, n (% ²)		
Calving issue ³	5 (15)	1 (8)
Metritis ⁴	3 (9)	2 (17)
Hyperketonemia ⁵	8 (24)	4 (33)

¹Percentages calculated from total number of cows.

²Percentages calculated from total in each Ca status group.

³Twins or stillbirth.

⁴Metritis defined as rectal temp $\geq 39.5^\circ\text{C}$ and presence of fetid, red-brown, watery uterine discharge.

⁵Blood BHB > 1.2 mmol/L within 7 DIM.

Patterns of Intake, Rumination, Calcium, Systemic Inflammation, and Milk Production

Summaries of the final models used to assess pre- and postpartum patterns of physiology are presented in Tables

4 and 5, respectively.

DMI. Patterns of DMI measured daily prepartum and every 24 h from calving to 14 DIM postpartum are depicted in Figure 1A. Over the 14 d leading up to parturition, daily DMI did not differ between cows that would go on to experience dyscalcemia and cows that would be eucalcemic postpartum (Figure 1A). In the first 24 h postpartum, mean intake of DYS and EU cows were comparable (DYS: 17.2 kg/d, 95% CI = 15.4–19.0 kg/d; EU: 18.3 kg/d, 95% CI = 17.1–19.4 kg/d; $P > 0.1$); however, starting between 24 h and 48 h postpartum, DYS cows began eating less than EU cows (DYS: 16.3 kg/d, 95% CI = 14.5–18.1 kg/d; EU: 20.6 kg/d, 95% CI = 19.5–21.7 kg/d; $P = 0.004$). Dyscalcemic cows also ate less than EU cows between 72 and 96 h postpartum (DYS: 18.8 kg/d, 95% CI = 17.0–20.6 kg/d; EU: 23.3 kg/d, 95% CI = 22.1–24.4 kg/d; $P = 0.004$), and between 96 and 120 h postpartum (DYS: 19.6 kg/d, 95% CI = 17.8–21.4 kg/d; EU: 22.9 kg/d, 95% CI = 21.8–24.0 kg/d; $P = 0.05$).

Rumination Time. Periparturient patterns of RT from 14 d prepartum to 8 h prepartum, and from the time of parturition through 336 h postpartum, are depicted in Figure 1B. Rumination of cows during the prepartum period did not differ statistically between cows classified as dyscalcemic or eucalcemic during the postpartum period (Figure 1B). Postpartum, DYS and EU cows did not differ during the first 24 h with respect to RT. Dyscalcemic cows experienced depressed RT beginning at 40 h (DYS: 25.4 min/h, 95% CI = 19.3–31.4 min/h; EU: 38.2 min/h,

Table 3. Descriptive statistics (means $\pm$ SD, and medians with interquartile ranges [IQR]) for continuous variables measured before and after parturition in 45 multiparous Holstein cows by calcium status group; calcium status group classification was based on total serum calcium (tCa) at 96 h postpartum (dyscalcemic, DYS: tCa ≤ 2.2 mmol/L; eucalcemic, EU: tCa > 2.2 mmol/L)

Variable	Calcium status group			
	EU		DYS	
	Mean ($\pm$ SD)	Median (IQR)	Mean ($\pm$ SD)	Median (IQR)
Calf weight, kg	42.1 (± 6.1)	41.1 (36.0–47.6)	42.8 (± 4.0)	41.6 (40.3–43.8)
p305ME, ¹ $\times 10^3$ kg	14.3 (± 1.9)	14.0 (13.1–14.9)	14.4 (± 1.9)	14.4 (13.6–15.5)
Milk yield through 336 h, kg/milking	13.2 (± 3.7)	13.9 (11.4–15.7)	12.0 (± 3.7)	12.6 (10.3–14.6)
Milk yield through 10 wk, kg/d	49.1 (± 8.9)	49.8 (43.1–54.7)	45.5 (± 8.6)	46.1 (39.8–52.7)
DMI, kg/d				
Prepartum	15.4 (± 2.6)	15.4 (13.6–17.3)	15.1 (± 2.7)	14.9 (13.4–16.8)
Postpartum	22.3 (± 3.4)	22.3 (20.2–24.7)	20.2 (± 4.0)	20.1 (18.4–22.6)
Rumination time, min/h				
Prepartum	43.7 (± 10.2)	43.8 (36.8–51.0)	41.9 (± 11.4)	41.5 (34.6–49.3)
Postpartum	43.6 (± 10.9)	44.3 (37.8–50.5)	37.4 (± 13.0)	39.3 (29.7–46.3)
tCa, mmol/L				
Prepartum	2.44 (± 0.09)	2.45 (2.37–2.50)	2.42 (± 0.08)	2.42 (2.37–2.50)
Postpartum	2.21 (± 0.23)	2.25 (2.07–2.35)	2.11 (± 0.25)	2.15 (2.00–2.27)
Haptoglobin, g/L				
Prepartum	0.25 (± 0.09)	0.21 (0.20–0.28)	0.23 (± 0.09)	0.21 (0.20–0.24)
Postpartum	0.42 (± 0.33)	0.31 (0.22–0.50)	0.72 (± 0.77)	0.40 (0.26–0.76)
BHB, mmol/L				
Prepartum	0.6 (± 0.1)	0.5 (0.4–0.6)	0.5 (± 0.1)	0.5 (0.4–0.6)
Postpartum	0.9 (± 0.3)	0.9 (0.7–1.0)	0.9 (± 0.3)	0.9 (0.7–1.0)

¹Previous-lactation 305-d mature-equivalent milk production.

Table 4. Summary of final model fit for linear mixed effects models used to assess associations between calcium group status based on serum total calcium (tCa) at 96 h postpartum (dyscalcemic: tCa $\leq$ 2.2 mmol/L; eucalcemic: tCa $>$ 2.2 mmol/L) and various prepartum response variables over time, adjusting for relevant confounders

Prepartum response	Covariate, <i>P</i> -value							
	Group ¹	Time ²	Parity ³	TOC ⁴	p305ME ⁵	Group $\times$ time	Parity $\times$ time	TOC $\times$ time
DMI	0.1	0.8	0.03	0.09	0.1	0.8	<0.001	<0.001
Rumination	0.5	<0.001	0.09	0.001	0.07	0.5	—	<0.001
tCa	0.1	0.003	0.03	—	0.2	0.3	—	—
Haptoglobin	0.9	0.6	0.8	0.09	0.2	0.95	—	0.05
BHB	0.3	0.01	0.96	—	0.4	0.5	—	—

¹Calcium status group.

²Day prepartum.

³Parity group (2, 3, or $\geq$ 4).

⁴Time of calving (morning, afternoon, or night).

⁵Previous-lactation 305-d mature-equivalent milk production.

95% CI = 34.5–42.0 min/h; P = 0.05) and sustained through 64 h postpartum (DYS: 26.5 min/h, 95% CI = 20.5–32.6 min/h; EU: 44.9 min/h, 95% CI = 41.2–48.7 min/h; P = 0.001), and again at 208 h postpartum (P = 0.05; Figure 1B).

Total Serum Calcium. Patterns of tCa concentrations measured pre- and postpartum are depicted in Figure 1C. Prepartum patterns of tCa concentrations did not differ between cows that would go on to experience dyscalcemia and cows that would be eucalcemic postpartum. From 8 h postpartum through 88 h postpartum, we found no significant differences between the tCa of EU and DYS cows, although the pattern over time for both groups followed a characteristic initial reduction for the first 24 h followed by a slow recovery beginning in the next 24 h (24–48 h postpartum; Figure 1C). At 96 h postpartum, a measurable difference was detected between the tCa of DYS and EU cows (DYS: 2.11 mmol/L, 95% CI = 2.02–2.21 mmol/L; EU: 2.36 mmol/L, 95% CI = 2.31–2.43 mmol/L; P = 0.003) whereby DYS cows had lower concentrations of tCa than EU cows. By 120 h postpartum, tCa did not differ between calcium status groups, both groups had recovered prepartum tCa concentrations, and groups did not differ in tCa at 168, 240, or 336 h postpartum.

Haptoglobin. Concentrations of haptoglobin over time by calcium status group are depicted in Figure 1D. Prepartum haptoglobin concentrations remained low and did not differ between groups (Figure 1D). Postpartum, haptoglobin concentrations began to rise immediately and were comparable between groups until 64 h postpartum (DYS: 0.69 g/L, 95% CI = 0.49–0.98 g/L; EU: 0.39 g/L, 95% CI = 0.32–0.48 g/L; P = 0.05). Haptoglobin concentrations of DYS cows continued to increase through 96 h postpartum (DYS: 0.90 g/L, 95% CI = 0.65–1.25 g/L; EU: 0.36 g/L, 95% CI = 0.29–0.43 g/L; P = 0.001), at which point haptoglobin concentrations of DYS cows

began to decline, reaching comparable concentrations to those of EU cows by 240 h postpartum (Figure 1D).

BHB. Patterns of BHB concentrations over time by calcium status group are shown in Figure 1E. Prepartum BHB concentrations were low and did not differ between groups (Figure 1E). Postpartum, BHB concentrations rose from the time of calving through 336 h postpartum, although they did not differ statistically between groups at any time point.

Milk Yield Through 14 DIM. The short-term pattern of milk production from parturition through 336 h postpartum is shown in Figure 2. Dyscalcemic cows produced statistically less milk per milking than EU cows at 2 milkings, 128 and 160 h postpartum. Despite not reaching statistical significance at individual milkings, on average over the first 336 h postpartum, differences in milk production between DYS and EU cows amounted to 1.4 kg (95% CI = 0.6–2.1 kg) less milk produced by DYS cows per milking (P = 0.001).

Long-Term Milk Yield. Average daily milk yield by week through the first 10 wk of lactation is depicted in Figure 3. Dyscalcemic cows produced less milk than EU cows during wk 1, 2, 6, 7, 8, and 9 of lactation (all $P \leq$ 0.05). On average, this amounted to 4.8 kg/d (95% CI = 2.3–7.4 kg/d) less milk produced by DYS cows through 10 wk (P < 0.001).

DISCUSSION

Our primary objective was to assess periparturient patterns of physiology to establish the sequence of measurable abnormalities occurring in association with dyscalcemia, a known form of postpartum maladaptation. We hypothesized that a decline in DMI followed shortly by an increase in inflammation would occur immediately after parturition, and that these 2 processes would be followed by reduced blood calcium concentra-

Table 5. Summary of final model fit for linear mixed effects models used to assess associations between calcium group status based on serum total calcium (tCa) at 96 h postpartum (dyscalcemic: tCa ≤ 2.2 mmol/L; eucalcemic: tCa > 2.2 mmol/L) and various postpartum response variables over time, adjusting for relevant confounders

Postpartum response	Covariate, <i>P</i> -value														
	Group ¹	Time ²	Parity ³	TOC ⁴	p305ME ⁵	Calf weight	Calving issue ⁶	Group × time	Parity × time	TOC × time	Calf weight × time	Calving issue × time	CH ⁷	FM ⁸	PG ⁹
DMI	0.002	<0.001	0.02	0.07	0.002	—	—	0.07	—	—	—	—	0.98	0.04	0.7
Rumination	0.4	<0.001	0.7	0.2	0.8	—	—	<0.001	—	<0.001	—	—	0.4	0.1	0.6
tCa	0.8	<0.001	0.5	0.1	0.2	—	—	<0.001	—	<0.001	—	—	0.5	0.09	0.8
Haptoglobin	0.3	<0.001	0.5	0.3	0.6	0.6	0.08	<0.001	—	—	<0.001	<0.001	0.1	<0.001	0.96
BHB	0.6	<0.001	0.8	0.004	0.6	—	—	0.4	—	<0.001	—	—	<0.001	0.4	0.1
Milk yield, 336 h	0.05	<0.001	0.01	0.2	<0.001	0.006	0.6	0.7	0.005	—	0.04	0.07	0.1	0.004	0.2
Milk yield, 10 wk	0.003	<0.001	0.05	0.5	<0.001	0.005	0.07	0.01	0.004	0.1	—	0.08	0.6	0.1	0.08

¹Calcium status group.

²Hour postpartum.

³Parity group (2, 3, or ≥ 4).

⁴Time of calving (morning, afternoon, or night).

⁵Previous-lactation 305-d mature-equivalent milk production.

⁶Twins or stillbirth.

⁷Treatment with ceftiofur hydrochloride.

⁸Treatment with flunixin meglumine.

⁹Treatment with propylene glycol.

tions, increased BHB, and depressed RT, followed by impaired productivity. Our findings with respect to this first objective did not wholly support our hypothesis and instead suggested that depressed RT occur concurrently with reduced DMI. We also determined that milk production deficits during the first 336 h postpartum are real but largely difficult to measure on a per-milking basis, and that BHB physiology may contribute minimally, if at all, to postpartum dyscalcemia. Our secondary objective was to support the premise that early postpartum dysregulation has longer-term relevance, and our findings firmly support our hypothesis that dyscalcemia would be associated with sustained milk production losses. This secondary finding highlights the importance of studying physiological dysregulation during the early postpartum period.

Regarding rumination and feed intake, our observations were somewhat unexpected. Rumination times have historically been an inconsistent predictor of DMI (Schirrmann et al., 2012; Clément et al., 2014; Souza et al., 2022), but both followed similar temporal patterns in this study. Schirrmann et al. (2012) and Clément et al. (2014) conducted their studies using mid-lactation cows, and the relationship between rumination and feed intake may depend on stage of lactation, as many physiological relationships do. Although both intake and rumination have previously been shown to be impaired in dyscalcemic cows (Seely et al., 2021a; Seely and McArt, 2024), our findings demonstrate that these changes begin between 24 and 48 h postpartum, and that these changes occur concurrently.

Through a large body of literature implementing feed restriction, it has been well known for decades that decreased intake is causally upstream of impaired milk production (McGuire et al., 1995; Kvidera et al., 2017; Leduc et al., 2021). Overall, average milk production per milking was reduced for DYS cows with impaired intake during the first 336 h postpartum. This finding is in line with previously described causal structures, despite difficulties measuring these milk production differences on a per-milking basis. Noncausal associations have shown a similar link between increased rumination times and improved milk production (Johnston and DeVries, 2018; Kaufman et al., 2018; Souza et al., 2022), and our data also support these previous findings. In addition, our longer-term data support previous research suggesting that early postpartum deficits in rumination times are associated with poor peak milk yields and sustained productivity losses in multiparous cows (Peiter et al., 2021). Though it is impossible to be certain with this study design that direct causality exists between depressed intake and rumination in the early postpartum period and long-term production issues, the timing of events makes this hypothesis somewhat more plausible.

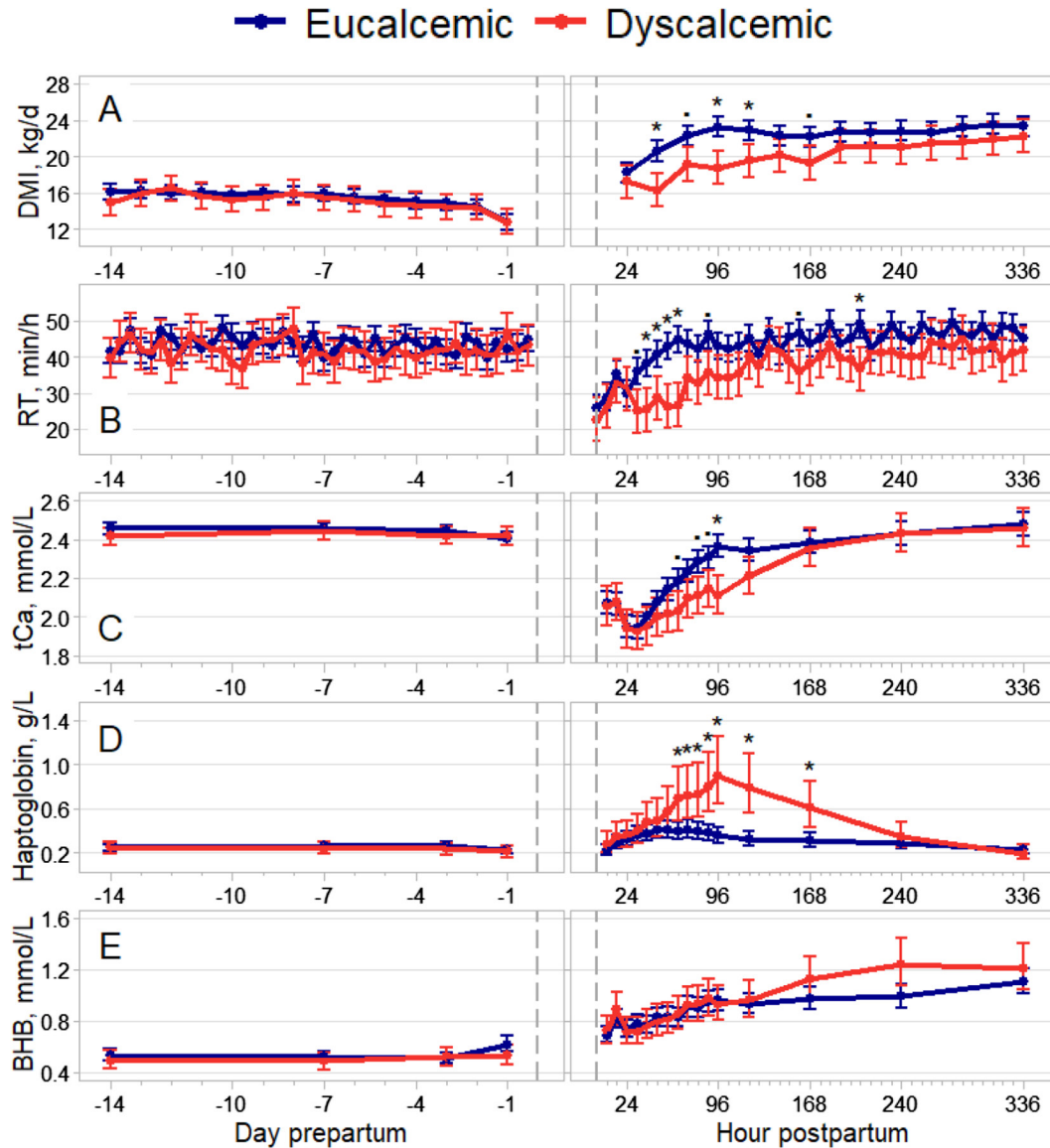


Figure 1. Temporal patterns of DMI (kg/d; A), rumination times (RT, min/h; B), serum total calcium (tCa, mmol/L; C), serum haptoglobin concentrations (g/L; D), and serum BHB (mmol/L; E) from 14 d prepartum to 336 h postpartum (14 DIM) by calcium status group for 45 multiparous Holsteins at the Cornell University Ruminant Center in Harford, New York. Calcium status group classification was based on tCa at 96 h postpartum (dyscalcemic: tCa ≤ 2.2 mmol/L, $n = 12$; eucalcemic: tCa > 2.2 mmol/L, $n = 33$). Pairwise hypothesis tests adjusted for multiple comparisons using the Benjamini-Yekutieli method. Dashed vertical lines indicate time of parturition. Dots (•) indicate differences between groups at $0.1 \geq P > 0.05$, and stars (*) indicate differences between groups at $P \leq 0.05$. Error bars represent 95% CI calculated from SE of predicted means.

It is conceptually easy to imagine how depressed intake could lead to low blood calcium concentrations; however, results of feed restriction experiments evaluating blood calcium concentrations in dairy cows tend to suggest that this treatment does not have a strong influence on tCa (Kume et al., 1998; de Sales Silva et al., 2020; Drackley et al., 2025). In our data, we found that a slower increase in intake preceded measurable tCa deficits, suggesting that during the early postpartum period a causal relationship is plausible. In contrast, Martinez et al. (2014) demonstrated that induced subclinical

hypocalcemia (defined as ionized Ca between 0.7 and 0.9 mmol/L) maintained for 24 h directly caused intake depression in nonpregnant nonlactating cows. Their work has been commonly cited to support the premise that subclinical hypocalcemia in early postpartum dairy cows has negative consequences on appetite and feeding behavior. Comparison here is challenging because we did not measure ionized calcium and, further, cannot establish any degree of causality; however, our data do not appear to support that sequence of events, at least during the early postpartum period.

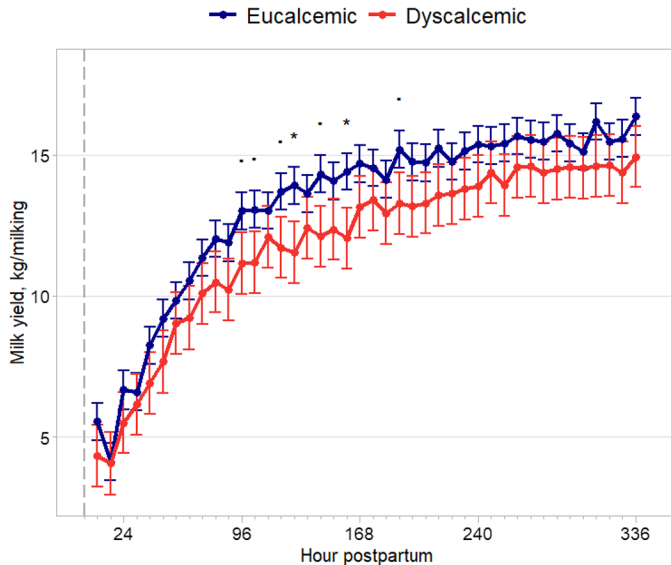


Figure 2. Temporal patterns of milk yield from calving to 336 h postpartum (14 DIM) collected every 8 h, by calcium status group, for 45 multiparous Holsteins at the Cornell University Ruminant Center in Harford, New York. Calcium status group classification was based on serum total calcium (tCa) at 96 h postpartum (4 DIM; dyscalcemic: tCa ≤ 2.2 mmol/L, n = 12; eucalcemic: tCa > 2.2 mmol/L, n = 33). Pairwise hypothesis tests adjusted for multiple comparisons using the Benjamini-Yekutieli method. Dashed vertical lines indicate time of parturition. Dots (•) indicate differences between groups at $0.1 \geq P > 0.05$, and stars (*) indicate differences between groups at $P < 0.05$. Error bars represent 95% CI calculated from SE of predicted means.

Early postpartum inflammatory activation has been commonly proposed as a potential initiator of maladaptation in this system (LeBlanc, 2012; Bradford et al., 2015; Horst et al., 2021). Inflamed cows eat less than noninflamed cows (Trevisi et al., 2015), induced inflammatory models suppress intake (Opgenorth et al., 2024b; Rodriguez-Jimenez et al., 2025; Westhoff et al., 2025), and the cytokine mediated mechanism that drives this relationship is solidly defined (Yuan et al., 2013; Trevisi et al., 2015; Ning et al., 2018). However, feed restriction in and of itself induces acute-phase inflammatory responses (Marques et al., 2012; Kvidera et al., 2017; Marques et al., 2019), and reductions in DMI precede the onset of inflammatory disease events (Bareille et al., 2003; Huzzey et al., 2007; Pérez-Báez et al., 2019). Combined, this evidence suggests that causality in the relationship between intake and inflammation can go in both directions.

Following an acute inflammatory insult, haptoglobin typically reaches peak concentrations within 48 h (Chandler et al., 2022b; Opgenorth et al., 2024a). Chronic inflammatory insults can delay the peak in haptoglobin concentrations up to 3 d after typical responses (Rodriguez-Jimenez et al., 2025). In our analysis, we found that haptoglobin concentrations began to diverge between

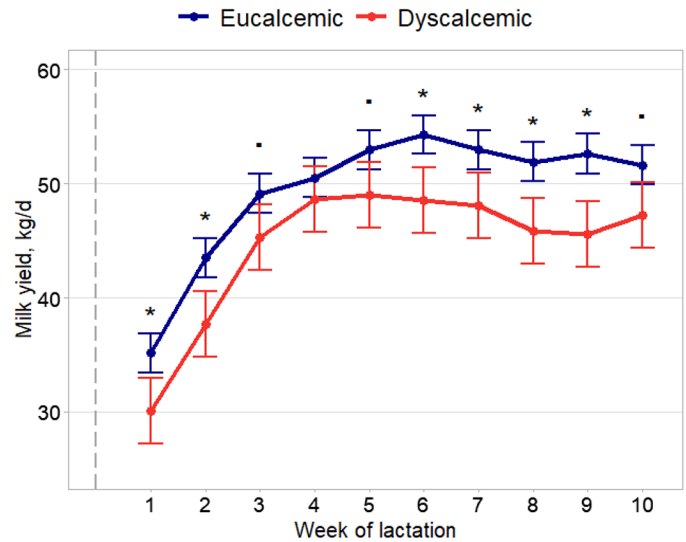


Figure 3. Average daily milk yield by week for the first 10 wk of lactation by calcium status group, for 45 multiparous Holsteins at the Cornell University Ruminant Center in Harford, NY. Calcium status group classification was based on serum total calcium (tCa) at 96 h postpartum (4 DIM; dyscalcemic: tCa ≤ 2.2 mmol/L, n = 12; eucalcemic: tCa > 2.2 mmol/L, n = 33). Pairwise hypothesis tests adjusted for multiple comparisons using the Benjamini-Yekutieli method. Dashed vertical lines indicate time of parturition. Dots (•) indicate differences between groups at $0.1 \geq P > 0.05$, and stars (*) indicate differences between groups at $P < 0.05$. Error bars represent 95% CI calculated from SE of predicted means.

DYS and EU cows at 48 h postpartum and reached a peak at 96 h postpartum. From this finding, and the previous data already described, it is plausible that the inflammatory insult sustained by these cows was chronic in nature and began around the time of parturition.

Criticisms of the traditional paradigm that aberrant physiology surrounding parturition is at the root of postpartum maladaptation come on the heels of a growing body of literature using induced inflammatory models, most frequently the LPS challenge model (Al-Qaisi et al., 2020; Horst et al., 2020; Horst et al., 2021). Despite the paucity of data collected under naturally occurring early postpartum conditions, these critiques have gained traction and reframed transition period thinking (Mezzetti et al., 2021; Lean et al., 2023; Eduardo Rico and Barrientos-Blanco, 2024). One of the key underlying motivators for our study was to assess the degree to which this LPS challenge evidence matches the natural physiological conditions occurring during the early postpartum period.

The initial cascade of physiology following LPS challenge is superficially similar to the physiological changes that we have uncovered in our data. Following the challenge, calcium concentrations in the blood decrease almost immediately, as do rumination times, feed intake, and milk production, followed by acute-phase re-

sponses, ~12 to 24 h behind (Al-Qaisi et al., 2020; Chandler et al., 2022a,b; Ogenorth et al., 2024a,b). After this similar initiation sequence, the postpartum physiological changes associated with dyscalcemia and LPS challenge responses diverge in 3 important areas: duration, magnitude, and associated outcomes.

In terms of duration, the LPS challenge consistently induces short-lived (≤ 24 h) reductions in DMI, rumination, milk production, and calcium concentrations (Al-Qaisi et al., 2020; Chandler et al., 2022a; Ogenorth et al., 2024b). Attempts to prolong inflammation with repeated LPS administration have generated a “sawtooth” pattern that is not consistent with the more persistent intake deficits we observed (Rodriguez-Jimenez et al., 2025). With respect to magnitude, haptoglobin concentrations consistently peak below 0.5 g/L in LPS challenge studies (Ogenorth et al., 2024a; Rodriguez-Jimenez et al., 2025), whereas the peak of haptoglobin concentrations in our cohort reached 0.90 g/L, and has been shown to reach even higher concentrations in commercial settings (~1.6 g/L; Seminara et al., 2025; Huzzey et al., 2009). Finally, LPS challenges do not induce prolonged productivity losses (Ogenorth et al., 2024b; Rodriguez-Jimenez et al., 2025; Westhoff et al., 2025), whereas early postpartum maladaptive phenotypes are associated with sustained milk yield depression. These important discrepancies clearly underscore the reality that physiology associated with an LPS challenge does not accurately recapitulate naturally occurring early postpartum maladaptive phenotypes. Our findings further suggest that the traditional postpartum paradigm may continue to be relevant despite the ideological challenges mounted against it.

The most critical limitation of our study relates to its external validity. To isolate the physiology associated with the maladaptive syndrome of dyscalcemia, we conducted our study in a controlled research farm environment, a choice that likely sacrificed some of the broader generalizability of our findings. By housing cows in individual tiestalls, we were able to measure individual intakes and remove the potential influence of social pressures; however, it is highly probable that, in a commercial setting, social factors contribute to the physiological processes that we have reported. By employing an intensive feed management program, whereby postpartum cows were offered fresh feed immediately after calving and 3 times daily, we were able to precisely measure intakes during key timeframes and ensure that the effects we observed were not due to feed availability. Although this was a necessary step to ensure that our intake response would be physiologically accurate, this type of feeding program is unrealistic and does not reflect management on a commercial dairy farm. It is more than likely that management factors, including how quickly fresh cows are offered feed after calving and how often feed is

pushed up or delivered, have effects on the underlying physiology, potentially driving further dysregulation that we were unable to observe. It is also possible that this aggressive feed management strategy and tiestall housing artificially deflated our incidence of dyscalcemia by shielding cows from social and management factors that could ultimately be contributing to the onset of postpartum dysregulation.

The sample size, though adequate to detect many of the differences we expected, was small and therefore may have been underpowered to reveal differences in early postpartum milk production and BHB physiology, as well as differences in our other responses that may have been occurring at other time points. Information bias around disease diagnoses is also a possibility, given that no veterinarian was involved in establishing disease presence or absence. Although the potential effect of this bias is somewhat minimal given the small proportions of animals developing disease, if it occurred, it is unclear how such a bias may have influenced results. Furthermore, the fact that this study was only conducted on a single farm, in a single geographic region, during a single season, also limits extrapolation to the general dairy cow population.

It is not clear to what extent the patterns we have reported are at play in commercial environments, but we can confidently state that under controlled conditions, a substantial proportion of multiparous Holstein dairy cows experience extensive physiological dysregulation in the early postpartum period that can be identified through the diagnosis of dyscalcemia. Additionally, our data show that this maladaptive phenotype is associated with long-term impaired milk production and therefore has relevance for individual cow welfare and whole-farm economics. Our findings further indicate that this maladaptation begins shortly after parturition, not before, and that it involves multiple axes of physiology at once, including intake, rumination, calcium regulation, and inflammation. It was difficult to establish a clear sequence of events from our data, despite intensive sampling and fine temporal resolution, which suggests that each dysregulated process begins following a common insult, which may or may not be parturition itself. It will be the role of future scientists to elucidate whether and how the process of parturition contributes to this phenotype. Using this data as a framework, it may be possible to develop strategies for diagnosis and treatment of maladapted cows, as the early rumination decline that we have observed may have diagnostic relevance and could be used to inform therapeutic interventions. Seely and McArt (2024) showed that rumination and activity time from 1 to 4 DIM were 78% accurate to identify dyscalcemia, supporting the utility of these data. Although this represents a real possibility, establishing which types of

therapeutic intervention might be applicable in this context remains a ripe area for future discovery.

CONCLUSIONS

Early postpartum lactational maladaptation begins in earnest within the first 48 h following calving, with feeding-related abnormalities. This condition manifests as dyscalcemia and elevated haptoglobin concentrations at 96 h postpartum and is associated with substantial short- and long-term milk losses. Although the strict causality in this system remains elusive, our study has illustrated that dysregulated physiology occurs simultaneously across multiple systems. Using this information as a basis, diagnostics and therapy should be explored to ameliorate suboptimal outcomes associated with early postpartum maladaptation.

NOTES

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Nonstandard abbreviations used: CH = treatment with ceftiofur hydrochloride; DYS = dyscalcemic cows; EU = eucalcemic cows; FM = treatment with flunixin

meglumine; IQR = interquartile range; p305ME = previous-lactation 305-d mature-equivalent milk production; PG = treatment with propylene glycol; RT = rumination time, min/h; tCa = total serum calcium concentration; TOC = time of calving.

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